

BANA 5160 CAPSTONE · PROGRESS UPDATE

NYC Airbnb Fair-Price Model

Model definition and initial results

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Data: Inside Airbnb NYC · snapshot June 14, 2026 · 9,752 active listings

PROJECT SCOPE

What we are building

A fair-price estimation tool for existing Airbnb hosts, framed as a dashboard add-on rather than a market-entry study.

Questions the tool answers

- Is a listing's nightly price above or below its estimated fair value?
- How does it compare to genuinely similar listings?
- Should the price be adjusted up or down?
- Which amenity or capacity factors explain the difference?

Scope decisions

- Audience: current hosts, not prospective investors.
- Deliverable: a per-listing price estimate plus its main drivers.
- Uses only Inside Airbnb data already available for each listing.
- Interpretability is a stated requirement, not an afterthought.

MODEL DEFINITION

A log-linear OLS model of nightly price

Target: $\log(\text{nightly_price})$. Price skewness is 7.36; the log transform reduces it to 0.27, satisfying OLS assumptions.

Predictors

- Categorical (one-hot, drop-first): borough, room type, host tier
- Capacity: max guests, bedrooms, beds, bathrooms
- 7 review sub-scores (cleanliness, location, value, ...)
- 20 most frequent amenities as binary flags

Preprocessing

- Active-listing filter (price, availability, recent reviews): 30,259 → 9,752 rows
- Bed/bath counts parsed from text; nulls median-imputed within capacity group
- Amenities encoded as 0/1 flags
- Coefficients converted to % price effect via $(e^{\beta} - 1) \times 100$

VALIDATION

How the model was built and tested

An 80/20 hold-out plus 5-fold cross-validation; all metrics below are on data the fitted model did not see.

80 / 20

train / test split

\$113.22

MAE (held-out test)

0.243 / 0.486 R^2 — price / log scale**39.2%**

median absolute % error

What we tested

- Fixed 80/20 hold-out; 5-fold out-of-fold predictions give every listing a residual from a model not trained on it.
- OLS fit with HC3 heteroskedasticity-robust standard errors.
- Ridge regression (5-fold CV penalty) as a multicollinearity check on the correlated review scores, $R^2 = 0.243 / 0.486$, effectively identical to OLS, indicating a stable fit.

MODEL COMPARISON

Why log-linear OLS over a black box

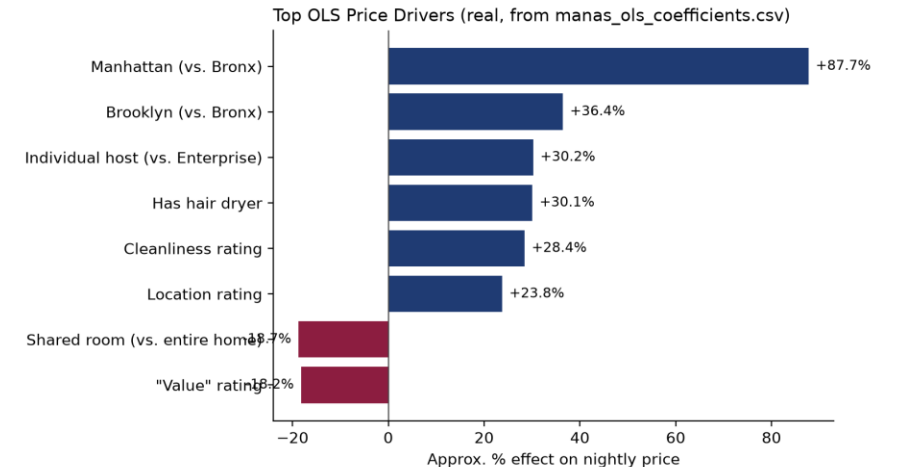
Random Forest is more accurate but not interpretable; OLS is retained as the product model, RF as an accuracy benchmark.

Model	MAE	R ² (price)	R ² (log)	Median APE
Log-linear OLS (chosen)	\$113.22	0.243	0.486	39.2%
Random Forest	\$92.94	0.506	0.619	29.3%

Why OLS is the product model

- Each coefficient reads directly as a % price effect.
- Stable under the price skew via the log transform + smearing correction.
- Guarantees positive price predictions.
- Produces the per-listing residual the tool depends on.

Estimated % effect on nightly price



Note: the negative sign on "has_hot_water" likely reflects the unresolved monthly-rental confound (see Next Steps).

KEY FINDINGS

What the initial results tell us

A first read of the fitted model, the data it exposed, and the limits we still need to address.

Location dominates price

Borough is the single largest driver, a Manhattan location is associated with roughly a **+88%** effect on nightly price, ahead of room type and capacity.

Review scores add modest signal

Cleanliness and location ratings carry measurable positive effects, but the seven review scores are correlated, hence the Ridge robustness check.

A confound remains

Several amenity coefficients (kitchen, hot water) carry unexpected negative signs, likely driven by monthly-rental listings mixed into the sample.

Snapshot

9,752

active listings modeled

\$113.22

mean absolute error (test)

48.6%variance explained (log R^2)**39.2%**

median absolute % error

SEGMENTS

A closer look at host tiers

The location and capacity effects above vary systematically by host tier, a segmentation we will exploit in the next revision.

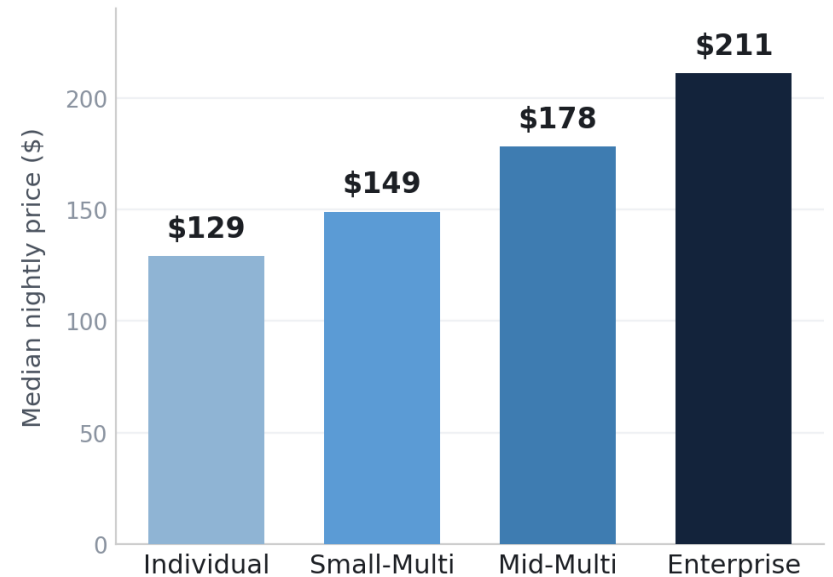
From the findings to segments

Median nightly price rises steadily from individual hosts (\$129) to enterprise operators (\$211), a ~64% spread. Host tier therefore carries pricing signal beyond location, capacity alone, and interacts with them.

Planned for the next revision

Comparable-listing layer (KNN) retrieving the nearest listings within a host-tier × borough × room-type segment, plus an **optimization / simulation layer** that searches for the revenue-maximizing nightly price and simulates the outcome against comparable listings.

Median nightly price by host tier



Synthetic sample (800 rows) matching the active_listings_clean_v6.csv schema; used to prototype the segmentation ahead of the next-rev build.

NEXT STEPS

Planned work before the next check-in

1

Separate long-stay listings

Monthly rentals (58.9%, min_nights > 28) average \$162.73/night vs. \$376.82 for short stays. Modeling them separately should reduce coefficient bias.

2

Resolve the negative amenity signs

Re-examine kitchen and hot-water coefficients; we expect the long-stay split above to resolve most of the confound.

3

Build the comparable-listing layer

KNN retrieval with exact segment match on borough × room type × capacity tier, and a standardized KNN fallback for thin segments.

4

Add review sentiment as a feature

Incorporate the BERT sentiment score (already computed on review text) into the model pipeline.